README

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Overview

An Optimal Sample-Selection System is a lightweight tool for generating optimal sample groupings.

- **Backend**: Flask (server.py) calls main.multi_round_greedy()
- **Frontend**: single-page HTML/CSS/JS GUI
- **Persistence**: save results as .db files; list / view / delete anytime

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Prerequisites

Tool	Version	Notes
Python	≥ 3.8	required
pip	latest	package installer
Flask	≥ 2.3	pip install flask flask-cors
Browser	modern	Edge / Chrome / Firefox

Quick Start

git clone https://github.com/Brandon030722/An_Optimal_Samples_Selection_System.git

NOTE: This project has been set up as a private project for the time being due to the MUST Artificial Intelligence instructor, and cannot be cloned.

```
cd    An_Optimal_Samples_Selection_System
pip install flask flask-cors
python server.py        # runs on http://127.0.0.1:8000
```

Open the URL above in your browser and you're ready.

Project Layout

```
project/
├─ server.py          # Flask backend
├─ main.py              # core algorithm (not shown here)
├─ static/
│    └─ index.html      # single-page frontend
└─ db/                    # auto-created storage for .db files
```

Front-End Walk-Through

#	UI Element	Purpose
1	**m**	total pool size
2	**n**	number of samples to pick
3	**Mode**	**Random n** or **Selected n**
4	**Selected IDs**	enabled only in *Selected n* mode
5	**k / j / s**	algorithm parameters
6	**time_limit**	Maximum number of seconds of operation
7	**min_s_covered(**Advanced parameters)	min coverage per sample
8	**max_trials(**Advanced parameters)	outer loop cap
9	**sample_per_round**	samples per greedy round
10	**Run**	start algorithm (shows progress bar)
11	**Store (.db)**	persist result to db/
12	**Go to DB**	switch to DB page
13	**Clear**	reset to defaults
14	**Result**	shows Selected list and Gxx groups
15	**DB Page**	list/display/delete/print .db files

Parameter Reference

Name	Type	Default	Description
m	int	45	size of full sample pool
n	int	10	number of samples to choose
Mode	radio	Random	selection mode
selected	list[int]	–	explicit IDs in *Selected* mode
k	int	6	items per group
j	int	4	overlap constraint
s	int	4	target co- occurrence
min_s_covered	int	1	min coverage per item
max_trials	int	50	outer loop iterations
sample_per_round	int	200	candidates per round
time_limit	int	60	Maximum number of seconds of operation

Tooltips (?) provide English hints.

Typical Workflow

1. Fill **m, n, k, …**
2. Choose **Selected n** and enter IDs (**if needed)**
3. Click **Run** – watch the blue progress bar
4. If satisfied, click **Store (.db)**
5. Switch to **DB Page** → *Display* or *Delete* files or print (**if needed)**

Database (.db) Handling

Files live in `db/` with pattern `m-n-k-j-s-idx-groups.db`

Copy files to other systems for further analysis if you like

FAQ

Question	Fix
Run button seems dead	1) backend not running 2) bad parameters (check browser console)
CORS errors	open via 127.0.0.1:8000, not file://
Cannot write .db	ensure db/ exists and is writable
